```
    GPIO.output(FAN, GPIO.LOW)
    return "Fan OFF"


if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

7. Execute the Above Program

i) Run the above Program.

ii) Find Raspi IP Address, by executing below command on command prompt:

hostname -I

Eg: 192.168.1.10

iii) Open Web Interface on mobile or laptop and enter the IP address in browser.

Eg: http://192.168.1.10:5000

8. Result

The web dashboard using Flask is designed successfully to control the relay module via the web.

PRACTICAL – 15

1. Aim

Create two factor authentication system by interfacing fingerprint sensor, RFID reader with Raspberry Pi.

2. Prior Concepts

i) Concept of Python Programming.

ii) Basics of serial (UART) communication.

iii) Basics of SPI communication.

iv) Basics of I2C communication.

v) Understanding of biometric systems.

3. New Concepts

i) Designing a Two-Factor Authentication (2FA) system.

ii) Integrating multiple authentication devices in a single application.

iii) Displaying authentication status on I2C LCD.

4. Hardware Required

i) MFRC522 RFID Reader

ii) R307/AS608 Fingerprint Sensor

iii) 16×2 LCD with I2C (PCF8574)

iv) Jumper Wires

5. Procedure

Pin Connection

RFID Pin	**Raspberry Pi Pin**
SDA (SS)	GPIO8 (CE0)
SCK	GPIO11
MOSI	GPIO10
MISO	GPIO9
RST	GPIO25

GND	GND
3.3V	3.3V

Fingerprint	**Raspberry Pi**
VCC	5V
GND	GND
TX	GPIO15 (RXD)
RX	GPIO14 (TXD)

LCD	**Raspberry Pi**
VCC	5V
GND	GND
SDA	GPIO2
SCL	GPIO3

Input IMAGE

i) Install Python3 on Raspi.

Write the below code in command prompt:

sudo apt install idle3

Enable following Interfaces:

sudo raspi-config

SPI -> Enable

I2C -> Enable

UART -> Enable

sudo reboot

Install Libraries

sudo apt update
pip3 install pyfingerprint
pip3 install mfrc522
pip3 install RPLCD
pip3 install RPLCD-i2c
pip3 install spidev
pip3 install RPi.GPIO

6. Implementation with Python Code

Code:

```
from pyfingerprint.pyfingerprint import PyFingerprint
from mfrc522 import SimpleMFRC522
from RPLCD.i2c import CharLCD
import RPi.GPIO as GPIO
from time import sleep

# ------------- LCD -------------
lcd = CharLCD('PCF8574', 0x27)
```

```
# ---------- RFID ----------
reader = SimpleMFRC522()

AUTHORIZED_UID = 123456789012

# ---------- Buzzer ----------

BUZZER = 18

GPIO.setwarnings(False)
GPIO.setmode(GPIO.BCM)
GPIO.setup(BUZZER, GPIO.OUT)

# ---------- Fingerprint ----------

try:
    finger = PyFingerprint(
        '/dev/serial0',
        57600,
        0xFFFFFFFF,
        0x00000000
    )

    if not finger.verifyPassword():
        raise ValueError()

except Exception:
    print("Fingerprint Sensor Error")
    exit()

# ---------- Functions ----------

def buzzer_beep(times):
    for i in range(times):
```

```
GPIO.output(BUZZER, 1)

sleep(0.2)

GPIO.output(BUZZER, O)

sleep(0.2)

def Icd_msg(line1, line2=""):

Icd.clear()

Icd.write_string(line1)

if line2:

Icd.crif()

Icd.write_string(line2)

#--------------------Main Program-------------

while True:

Icd_msg("Scan RFID Card")

try:

uid, text = reader.read()

if uid == AUTHORIZED_UID:

Icd_msg("RFID Matched")

buzzer_beep(1)

sleep(2)

Icd_msg("Place Finger")
```

```
while not finger.readImage():
finger.convertImage(0x01)
result = finger.searchTemplate()
position Number = result[0]
if position Number >= 0:
lcd_msg("Fingerprint", "Matched")
buzzer_beep(2)
sleep(2)
lcd_msg("ACCESS", "GRANTED")
print("ACCESS GRANTED")
else:
lcd_msg("Fingerprint", "Unmatched")
buzzer_beep(4)
sleep(2)
lcd_msg("ACCESS", "DENIED")
print("Fingerprint Failed")
else:
lcd_msg("RFID", "Unmatched")
buzzer_beep(4)
sleep(2)
```

```
lcd_msg("ACCESS", "DENIED")

print("Unauthorized RFID")

except KeyboardInterrupt:

GPIO.cleanup()

Break
```

7. Result:

PRACTICAL – 16

1. Aim

Create a motion detector by interfacing camera and 16X2 display with Raspberry Pi, Print the appropriate message on the display.

2. Prior Concepts

i) Concept of Python Programming.

ii) Basics of camera operation.

iii) Basics of 12C communication.

iv) Interfacing 16x2 LCD with Raspberry Pi.

v) Basic image processing concepts.

3. New Concepts

i) Motion detection using computer vision.

ii) Raspberry Pi Camera interfacing.